

ALLAMA IQBAL OPEN UNIVERSITY
(Department of Mathematics)

WARNING

- 1. PLAGIARISM OR HIRING OF GHOST WRITER(S) FOR SOLVING THE ASSIGNMENTS WILL DEBAR THE STUDENT FROM THE AWARD OF A DEGREE/CERTIFICATE IF FOUND AT ANY STAGE.**
- 2. SUBMITTING ASSIGNMENT(S) BORROWED OR STOLEN FROM OTHER(S) ONES WILL BE PENLIZED AS DEFINED IN AIOU PLAGIARISM POLICY**

Course: Quantitative Reasoning-I (9424)

Semester: Autumn, 2026

Level: BS

Please read the following instructions for writing your assignments. (AD, BS, B.Ed, MA/MSc, Med, ODL Mode)

1. All questions are compulsory and carry equal marks, but within a question, the marks are distributed according to its requirements.
2. Read the question carefully and answer it according to the requirements of the question.
3. Avoid irrelevant discussion/information and reproducing from books, study guides, or allied material.
4. Handwritten scanned assignments are not acceptable.
5. Upload your typed (in Word or PDF format) assignments on or before the due date.
6. Your own analysis and synthesis will be appreciated.
7. Late assignments can't be uploaded to the LMS.
8. The students who attempt their assignments in Urdu/Arabic may upload a scanned copy of the handwritten assignments (in PDF format) on the University LMS. The size of the file should not exceed 5MB.

Total No. 100

Pass Marks: 50

Assignment No. 1

(Unit 1 to 5)

Question 1

(20)

- (a) A circle is divided into three equal sectors, and each sector is further divided into four equal parts. If the radius of the circle is 6 cm, determine the total area of one sector and one part, and then find the total area of the entire circle.

- (b) Mubeen invested Rs. 100,000 in a stock that grew at an average rate of 6% per year for 7 years. How much money will Mubeen have at the end of the 7 years, assuming the growth rate remains constant?

Question 2 **(20)**

- (a) The number of atoms in a typical grain of sand is estimated to be about 10^{18} . If there are approximately 7.5×10^9 people in the world, estimate how many grains of sand there are per person on Earth.
- (b) You have Rs. 500 to spend on purchasing tickets for a concert. General admission tickets cost Rs. 25 each, and VIP tickets cost Rs. 50 each. Write and solve an equation to determine how many tickets of each type you can afford to buy without exceeding your budget. Show all the steps of your solution.

Question 3 **(20)**

- (a) Sarah deposits Rs. 20,000 into a savings account that earns an annual interest rate of 4%, compounded monthly. If she does not make any withdrawals, how much money will be in her account after 5 years?
- (b) A swimming pool can be filled by a hose in 6 hours, while a drain can empty it in 8 hours. If both the hose and the drain are opened simultaneously, how long will it take to fill the pool? Express your answer in hours and minutes.

Question 4 **(20)**

- (a) Statement A: The value of $2^3 + 3^2$ is greater than the value of $3^3 + 2^2$.
Statement B: The value of $\sqrt{2} + \sqrt{3}$ is greater than the value of $\sqrt{2} + \sqrt{3}$.
Determine which statement is true and provide an explanation.
- (b) Explore the pattern in the sequence: 2, 5, 10, 17, 26, ... Analyze the pattern and predict the 10th term of the sequence.

Question 5 **(20)**

- (a) A company produces widgets with a specified weight of 100 grams. However, due to variations in the manufacturing process, each widget's weight has an uncertainty of ± 2 grams. If a customer receives a batch of

50 widgets, what is the total uncertainty in the combined weight of the widgets? Express your answer in grams.

- (b) Statement A: The volume of a cube with edge length 5 cm is greater than the volume of a sphere with radius 5 cm.
Statement B: The surface area of a cube with edge length 5 cm is greater than the surface area of a sphere with radius 5 cm.
Compare the validity of these two statements and explain your reasoning.

Total No. 100

Pass Marks: 50

Assignment No. 2 **(Unit 6 to 9)**

Question 1 **(20)**

- (a) Calculate the area of a trapezoid with bases 7 units and 12 units, and height 20 units.
(b) Find the total surface area of a cone with radius 9 units and slant height 20 units.

Question 2 **(20)**

- (a) State and prove De Morgan's law using a Venn Diagram.
(b) Solve the inequality $|7x - 13| < 5$ for x . Write the solution set in interval notation and illustrate it on a number line.

Question 3 **(20)**

- (a) Solve the inequality $x^2 - 4x + 2 < 0$, for x . Express the solution set in interval notation and sketch the solution on a graph.
(b) Consider the following statement: "If it is raining, then the streets are wet." Determine whether the following statements are true or false and justify your answer:
1. "If the streets are wet, then it is raining."
2. "If the streets are not wet, then it is not raining."

Question 4 **(20)**

Determine whether the following pairs of propositions are logically equivalent:

- a) $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$
- b) $p \rightarrow (q \wedge r)$ and $(p \rightarrow q) \wedge (p \rightarrow r)$
- c) $\sim(p \wedge q)$ and $(\sim p) \vee (\sim q)$

Question 5

(20)

- (a) A company manufactures two types of products, A and B. The profit from selling one unit of product A is Rs.3, and the profit from selling one unit of product B is Rs. 5. Due to limited resources, the company can produce at most 100 units of product A and 80 units of product B. Write a system of inequalities to represent the constraints on production and determine the maximum profit the company can achieve.
- (b) A survey of 150 shoppers revealed that 90 shoppers prefer shopping online, 80 shoppers prefer shopping in-store, and 50 shoppers prefer both online and in-store shopping. Using a Venn diagram:
 - 1) Illustrate the events of preferring online shopping (O) and preferring in-store shopping (I).
 - 2) Determine the number of shoppers who prefer either online or in-store shopping (or both).
 - 3) Find the percentage of shoppers who prefer online shopping but not in-store shopping.